Part-2

Spanning Tree Protocol (Toolkit)

- Topics covered:
    - Root Guard
    - Loop Guard

Root Guard (STP)

◆ Moving on to Root Guard

Now we're moving on to Root Guard.

Root Guard:

- Prevents a port from becoming a root port
- Disables the port if superior BPDUs are received
- Enforces the current root bridge

This is very useful when connecting your LAN to another LAN that you don't control.

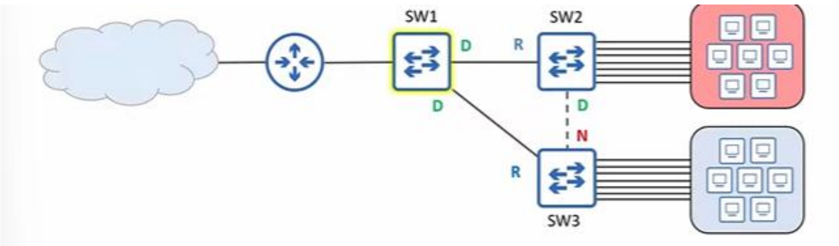
Another STP feature you need to know for CCNA is Loop Guard, which is covered next.

◆ Why keeping a specific root bridge is important

STP prevents loops by:

- Electing a root bridge
- Ensuring each switch has only one valid path to reach it

Initial LAN (SW1 as root bridge)



- SW1 is the root bridge
- SW2 has one active path: SW2 → SW1
- SW3 has one active path: SW3 → SW1
- SW2–SW3 link is disabled

Any switch could technically be root, but:

- You should not randomly select the root bridge

◆ Root bridge placement considerations

Two key considerations:

1. Optimal traffic flow
2. Stability and reliability

Optimal traffic flow

- Minimize latency
- Minimize congestion

Latency:

- Time it takes traffic to travel

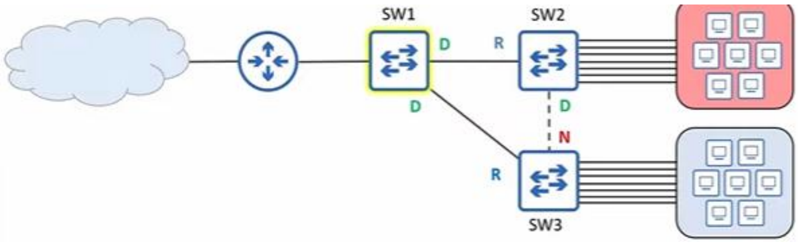
Congestion:

- How busy the network is

Too much traffic on a link:

- Frames wait
- Frames may be dropped
- Performance suffers

🔹 **Why SW1 was selected as root**



In most LANs:

- PCs don't communicate much with each other
- They communicate with servers outside the LAN
- Internet, WAN, etc.

So:

- PCs need an efficient path to R1
- SW1 connects to R1

Traffic paths:

- SW2 hosts → SW2 → SW1 → R1
- SW3 hosts → SW3 → SW1 → R1

This is the most efficient design.

🔹 **Stability and reliability**

After selecting a good root:

- You want it to remain stable

If you have:

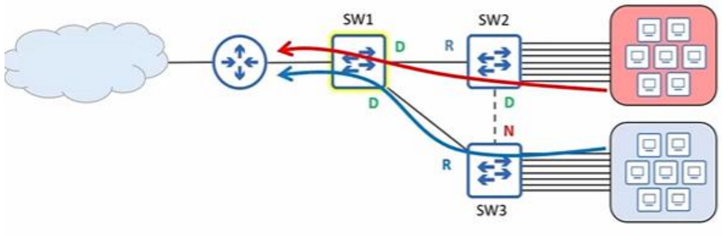
- New reliable switches
- Old unreliable switches

Then:

- Make the reliable switch the root bridge

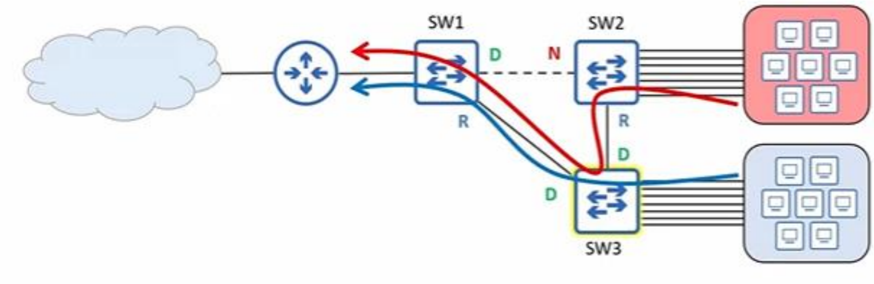
Root bridge considerations:

- Optimal traffic flow
- Stability and reliability



◆ **What if SW3 becomes the root?**

SW3 as root bridge:



Traffic impact:

- SW3 hosts → same path
- SW2 hosts → SW2 → SW3 → SW1 → R1

Problems:

- Extra hop
- Slightly more latency
- Possible congestion on SW3–SW1

Latency increase:

- Usually negligible

Main issue:

- **Congestion**
- Frames may queue or drop
- User experience suffers

Root bridge selection matters.

- **Expanding the network: service provider and customer**

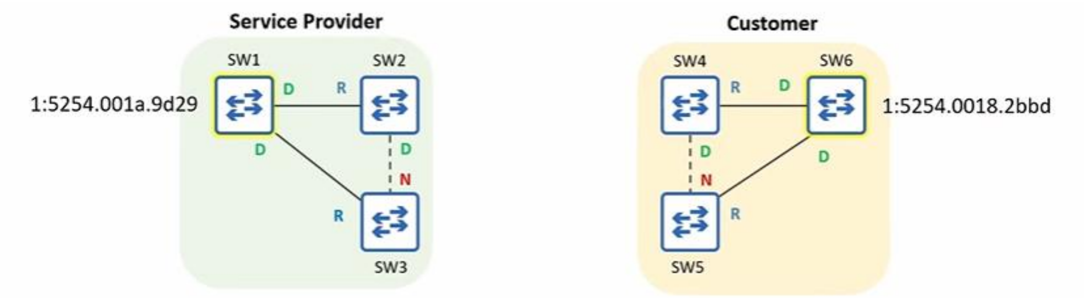
Within your own LAN:

- You can control the root bridge
- Set priority to 0 (plus VLAN ID)

Example:

- SW1 priority = 0 + VLAN ID → root
- SW6 priority = 0 + VLAN ID → root of its LAN

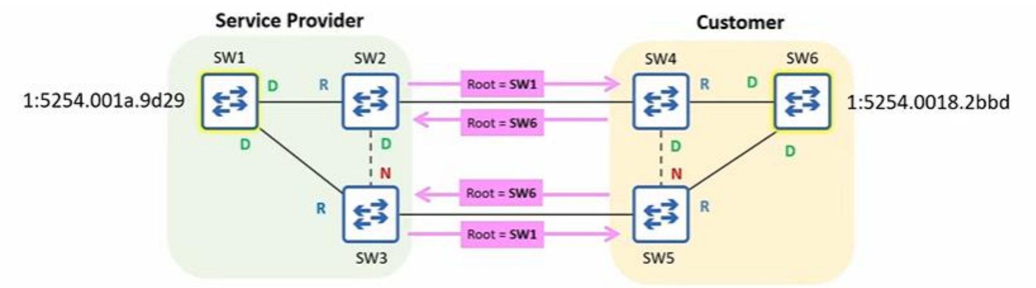
Two different organizations



Example:

- Service provider connects to customer
- Common in Metro Ethernet environments

What happens when the LANs connect?



- SW2 and SW3 send BPDUs saying SW1 is root
- SW4 and SW5 send BPDUs saying SW6 is root

Bridge ID comparison:

- Same priority

- Compare MAC addresses
- SW6 has lower MAC

Result:

- **SW6** becomes the root bridge

Lesson:

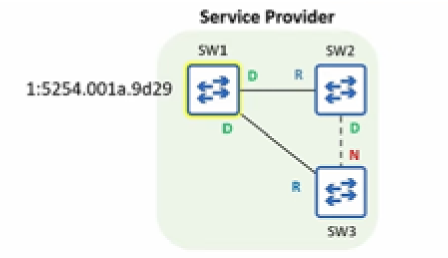
- Even priority 0 can lose
- Lower MAC address wins

🔹 Problem without Root Guard

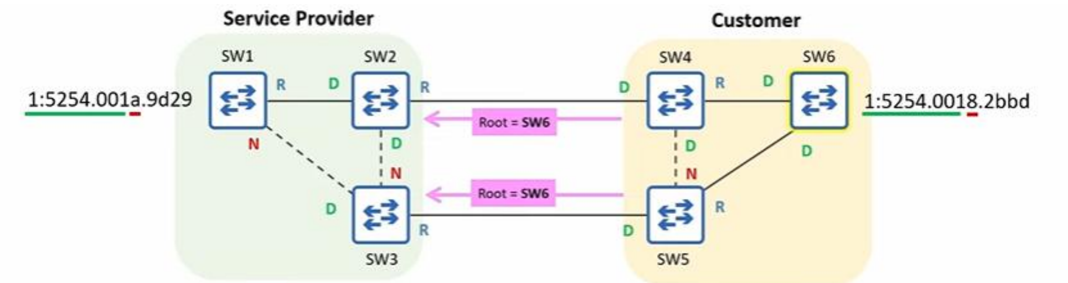
Without protection:

- SW1, SW2, SW3 accept SW6 as root
- Service provider topology changes

  Service provider topology before: -



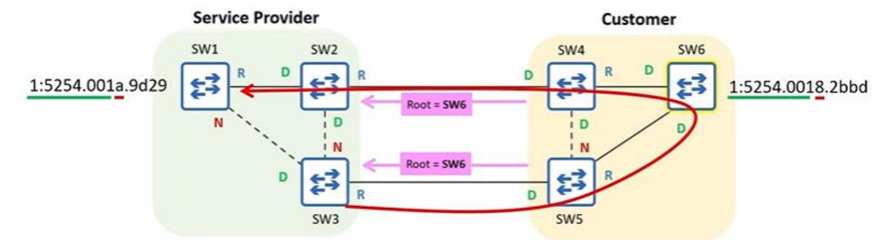
After:



Resulting topology:

- SW1–SW3 disabled
- SW2–SW3 disabled

Traffic path:



- Frames detour through customer LAN

This is:

- Inefficient
- Undesirable

◆ Root Guard solution

Root Guard:

- Prevents accepting superior BPDUs
- Enforces root bridge inside your LAN

Superior BPDU:

- Claims a better root bridge
- Lower bridge ID

SW6's BPDUs:

- Claim a better root than SW1
- Are superior

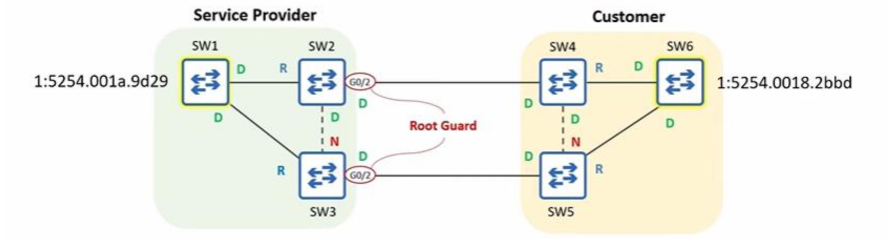
◆ Where to configure Root Guard

From service provider perspective:

- Ports connected to customer switches

Ports:

- SW2 G0/2
- SW3 G0/2



Command (interface config mode):

“SPANNING-TREE GUARD ROOT”

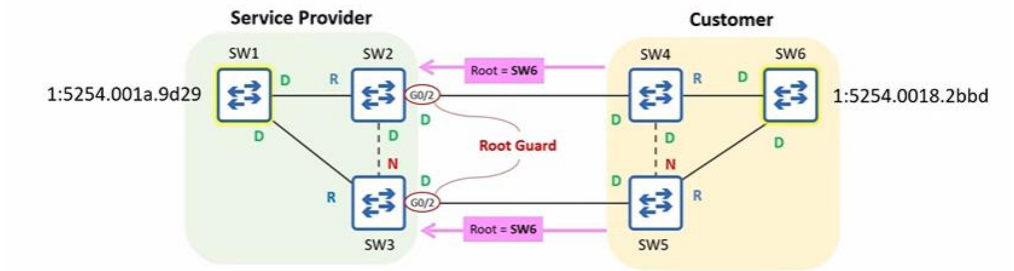
Important:

- No global default command
- Interface mode only

◆ **Root Guard behaviour**

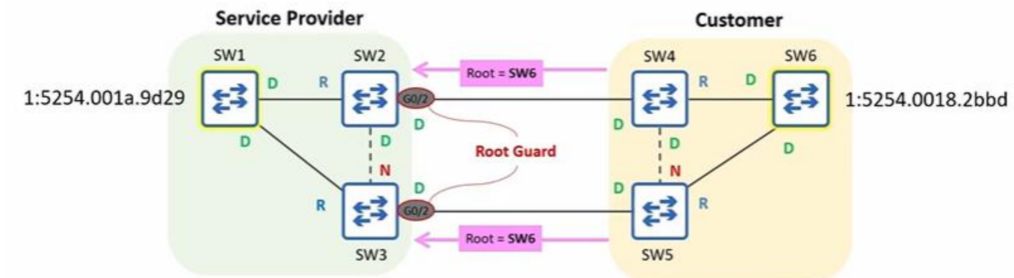
When a Root Guard–enabled port:

- Receives a superior BPDU



The port enters:

- Root-inconsistent
- Broken state



Effect:

- Port does not forward frames

- Frames are discarded
- All traffic is cut off

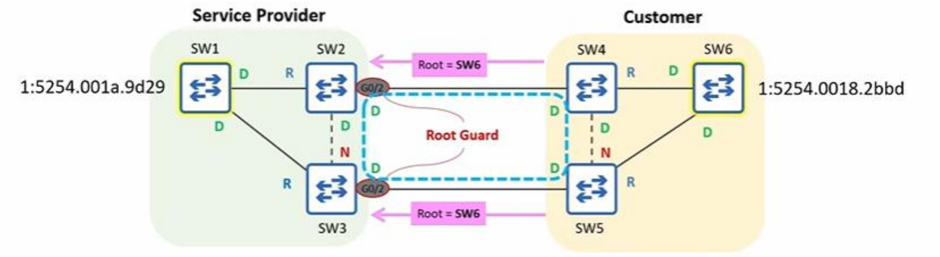
Result:

- SW1 remains root
- Customer LAN isolated

◆ Special note about designated ports

In this scenario:

- Both ends of the link are designated ports



Normally:

- Only one designated port per link

This is a special case:

- Caused by Root Guard blocking the link

◆ Restoring connectivity

Root Guard solved the provider's problem,
but now:

- Customer cannot communicate

To re-enable the port:

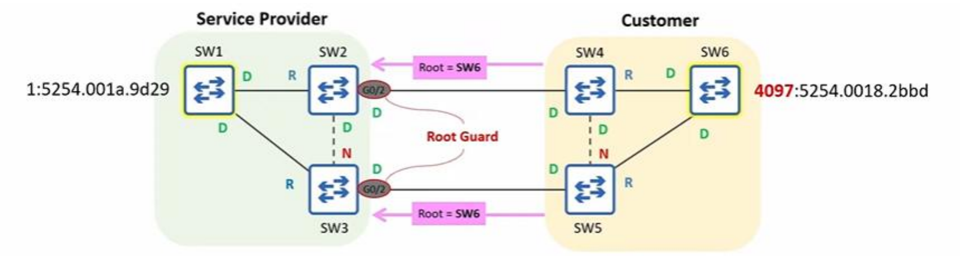
- The issue must be fixed
- Superior BPDUs must stop

Action:

- Service provider tells customer to increase priority

Customer change:

- SW6 priority = 4096 + VLAN ID



◆ **Automatic recovery**

After priority change:

- Superior BPDUs stop
- BPDUs age out

BPDU Max Age:

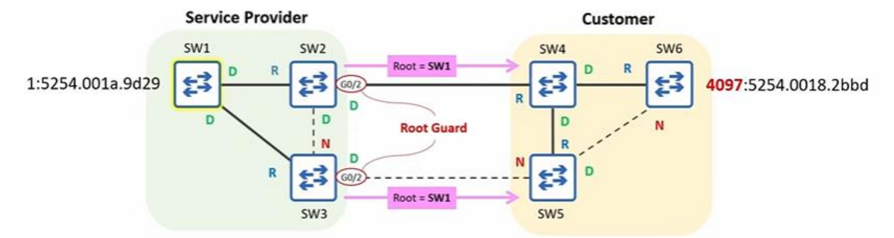
- 20 seconds (default)

After ~20 seconds:

- SW2 G0/2 re-enabled
- SW3 G0/2 re-enabled

No CLI action needed on SW2 or SW3.

◆ **Final topology after recovery**



- SW1 is root
- All switches have one active path to SW1
- Remaining links are blocked

◆ **Key point to remember**

To re-enable a port disabled by Root Guard:

- Do **not** configure anything on the Root Guard switch
- Stop the superior BPDUs
- Port recovers automatically

In this case:

- Customer modified bridge ID
- Problem resolved

.

◆ **Reviewing Root Guard in the CLI**

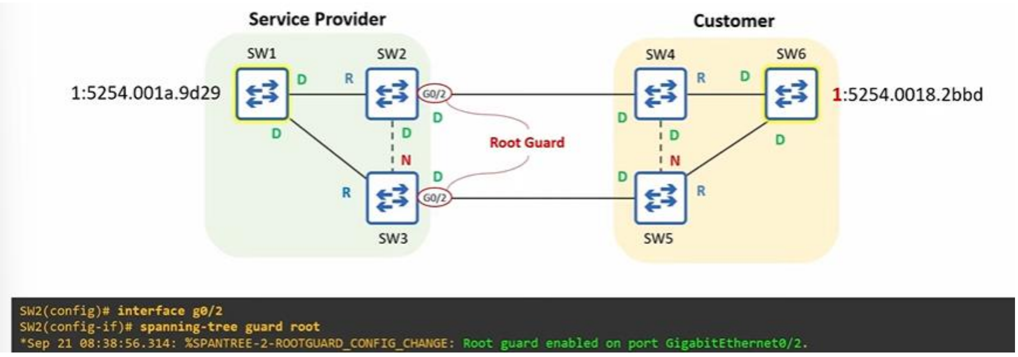
Root Guard only needs a single command to configure, so there is not much to show in the CLI.
Let's rewind and look at the process again.

At this point:

- SW6's STP priority is **1**
- That makes SW6's bridge ID **superior** to SW1's

Root Guard is enabled on:

- **SW2 G0/2**
- (Same configuration is also done on SW3 G0/2)



- **Root Guard configuration (interface mode)**

CLI (image shown once)

```
SW2(config)# interface g0/2
SW2(config-if)# spanning-tree guard root
*Sep 21 08:38:56.314: %SPANTREE-2-ROOTGUARD_CONFIG_CHANGE: Root guard enabled on port GigabitEthernet0/2.
```

SW2(config)# interface g0/2

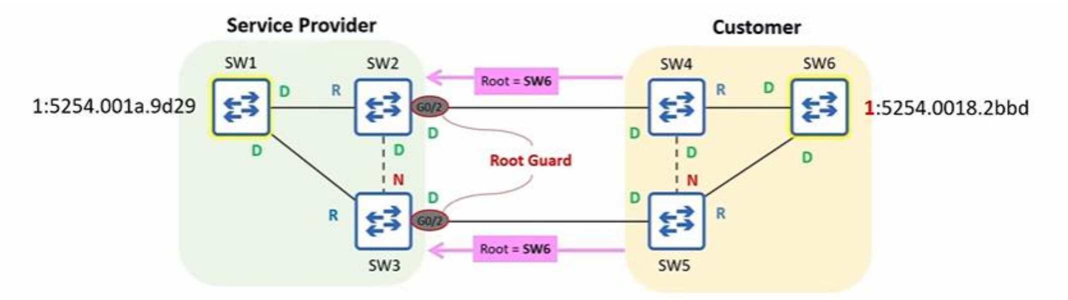
SW2(config-if)# spanning-tree guard root

After issuing the command:

- A message is shown
- Root Guard is enabled on the port

- **Root Guard blocking the port**

When SW2 and SW3 receive BPDUs from SW4 and SW5:



- Root Guard blocks the ports

```
SW2(config)# interface g0/2
SW2(config-if)# spanning-tree guard root
*Sep 21 08:38:56.314: %SPANTREE-2-ROOTGUARD_CONFIG_CHANGE: Root guard enabled on port GigabitEthernet0/2.
*Sep 21 08:38:56.321: %SPANTREE-2-ROOTGUARD_BLOCK: Root guard blocking port GigabitEthernet0/2 on VLAN0001.
```

CLI verification

"SHOW SPANNING-TREE"

```
SW2(config-if)# do show spanning-tree
!output omitted
Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/0               Root FWD 4         128.1    P2p
Gi0/1               Desg FWD 4         128.2    P2p
Gi0/2               Desg BKN*4         128.3    P2p *ROOT_Inc
```

BKN = Broken
ROOT_Inc = Root Inconsistent

Observed on G0/2:

- Status: **BKN**
- Right column: **ROOT_Inc**

Meaning:

- **BKN** = Broken
- **ROOT_Inc** = Root Inconsistent

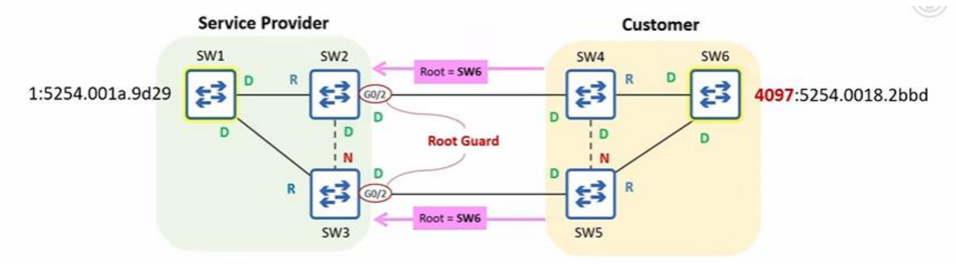
Explanation:

- A broken port is effectively disabled
- It cannot forward or receive data frames
- Root Inconsistent explains why it is broken
- The port was disabled by Root Guard

◆ Fixing the problem

To fix the issue:

- The customer increases SW6's priority



- New priority is **4097**

After this change:

- Superior BPDUs stop
- About 20 seconds later, the port recovers

CLI message

-
```
*Sep 21 08:54:26.955: %SPANTREE-2-ROOTGUARD_UNBLOCK: Root guard unblocking port GigabitEthernet0/2 on VLAN0001.
```

CLI verification

"SHOW SPANNING-TREE"

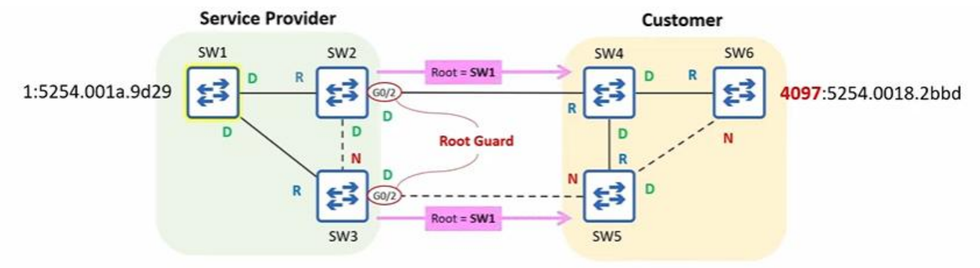
```
SW2(config-if)# do show spanning-tree
!output omitted
Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ------
Gi0/0               Root FWD 4         128.1    P2p
Gi0/1               Desg FWD 4         128.2    P2p
Gi0/2               Desg FWD 4         128.3    P2p
```

Observed:

- G0/2 is no longer broken
- Root Inconsistent message is gone

Network after recovery

The network converges again with SW1 as the root bridge.



Where Root Guard should be configured

- SW1, SW2, and SW3 belong to the service provider
- The service provider wants SW1 to remain the root bridge

Therefore:

- Root Guard is configured on ports connecting to customer switches
  - SW2 G0/2
  - SW3 G0/2

Important clarification:

- You should not configure Root Guard on every port connected to another network

Example:

- If the customer configures Root Guard on SW4 or SW5:
  - Those ports would be disabled when receiving superior BPDUs
  - Communication with the service provider would be blocked
  - That defeats the purpose of connecting to a provider

◆ **Root Guard summary**

- Root bridge selection should consider:
  - Optimal traffic flow
  - Minimizing latency and congestion
  - Stability and reliability
- Within your own LAN:
  - You can control the root bridge by setting priority to 0
  - If you control all switches, Root Guard is not needed
- Root Guard is useful when:
  - Connecting to switches outside your control
  - Such as service provider ↔ customer connections
- Root Guard:
  - Prevents a port from accepting **superior BPDUs**
  - Uses the command:
  - SPANNING-TREE GUARD ROOT
  - Can only be configured in **interface config mode**
  - Cannot be enabled by default globally
- When triggered:
  - Port becomes **broken**
  - Port is **root inconsistent**
  - Port cannot forward frames
- Recovery:
  - Happens **automatically**
  - No manual shutdown/no shutdown required
  - No ErrDisable Recovery required
  - Superior BPDUs must stop

That completes **Root Guard**.

Loop Guard (STP)

Moving on to Loop Guard

Now we're moving on to Loop Guard.

Loop Guard:

- Protects the network from loops
- Disables a port if it unexpectedly stops receiving BPDUs
- Ensures the port does not mistakenly enter the Forwarding state

This can happen due to:

- Software bugs
- Or, in this video's focus, a unidirectional link

Normal bidirectional link

Two switches are connected: **SW1** and **SW2**.

Under normal circumstances:

- SW1 can transmit data to SW2
- SW2 can transmit data to SW1

bidirectional

This is how links are supposed to work.

◆ **What is a unidirectional link?**

A unidirectional link:

- Data flows in only one direction

Example:

- SW1 → SW2 works
- SW2 → SW1 does not work

This is:

- Undesirable
- Usually caused by a Layer 1 issue

Possible causes:

- Damaged cables
- Faulty connectors
- Faulty transceivers (SFPs)

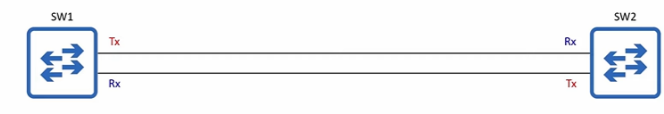
Unidirectional links are:

- More common with fibre-optic cables
- Less common with copper UTP cables

◆ **Why fiber links are more prone**

Fiber-optic connections:

- Use two separate fibers
  - One for Tx
  - One for Rx



Although separate:

- They are treated as one cable
- Both sides must work for full communication

Problem:

- If one fiber is damaged
- One direction can fail
- The other can still work

Fiber cables:

- Are more fragile than copper
- Bending too much can permanently break them

🔹 **Normal fiber behaviour (no unidirectional link)**

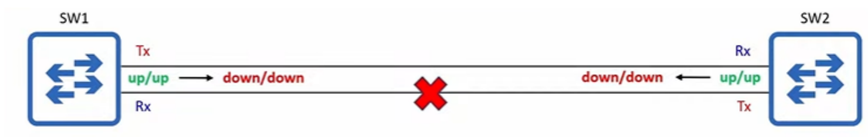
For a fiber interface to be **up/up**:

- Both fibers must work

Link status: up/up

If one fiber is cut or disconnected:

- Devices usually detect it
- Interfaces go down/down



Link status: down/down

In this case:

- No unidirectional link
- The link is completely disabled

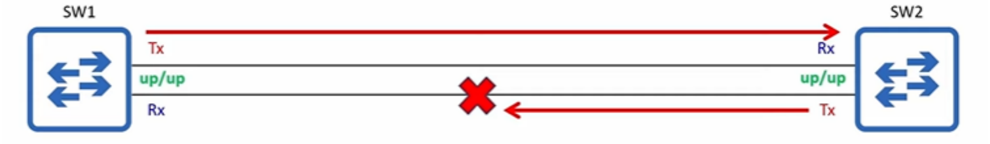
🔹 **When unidirectional links actually occur**

Sometimes:

- A physical issue is **not detected**
- Interfaces remain **up/up**

Result:

- One direction works
- The other does not



Link status: up/up

Both switches:

- Think the link is working
- But it only works one way

🔹 **Why unidirectional links are a problem for STP**

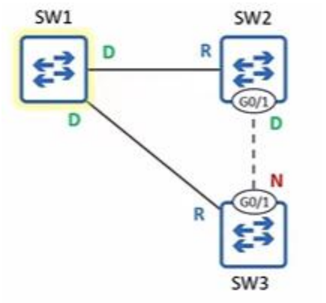
BPDUs:

- Originate from the **Root Bridge**
- Are sent every **2 seconds**
- Travel out **designated ports**

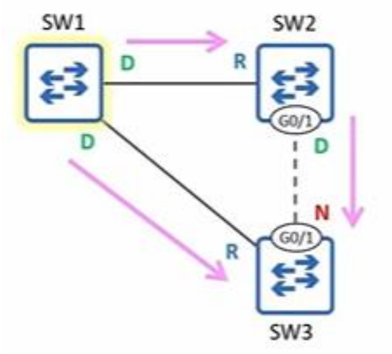
They are used to decide:

- Which ports forward
- Which ports block

◆ **STP example before the problem**



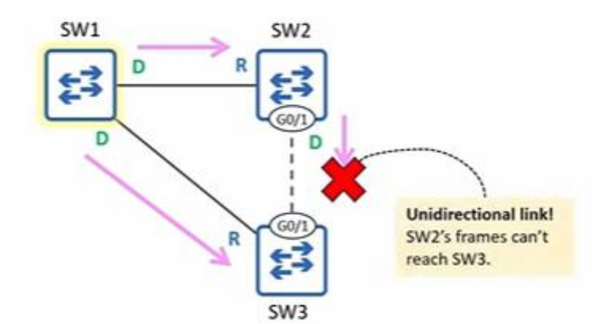
- SW3 G0/1 is a **non-designated blocking port**
- It receives **superior BPDUs from SW2**



◆ **Unidirectional link problem in STP**

Assume:

- SW2–SW3 link becomes unidirectional
- SW2's frames cannot reach SW3
- Interfaces remain up/up

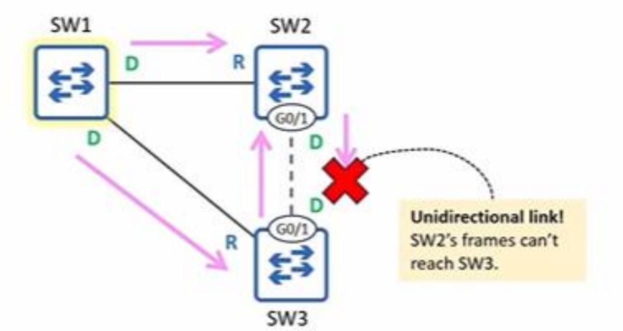


SW3 behaviour:

- Stops receiving BPDUs from SW2
- Assumes there is no loop

- Max Age timer counts down

After Max Age expires:



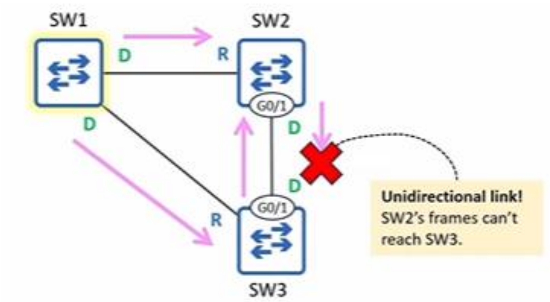
- SW3 G0/1 becomes **Designated**
- Starts forwarding BPDUs to SW2

But:

- SW3's BPDUs are inferior
- SW2 ignores them

Result:

- SW2 G0/1 = Forwarding
- SW3 G0/1 = Forwarding



A loop is created.

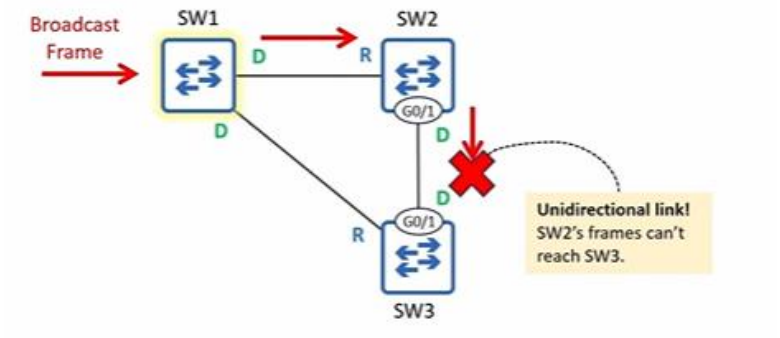
◆ **How the loop happens**

If SW1 receives a broadcast frame:

- It floods toward SW2 and SW3

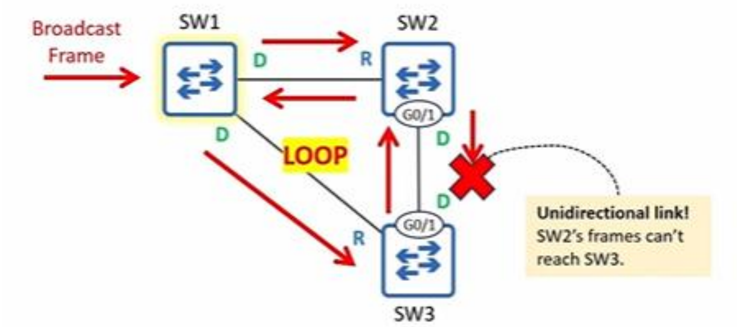
Frame behaviour:

- Frame sent to SW2 does not loop (unidirectional)



- Frame sent to SW3 loops endlessly

SW1 → SW3 → SW2 → SW1 → SW3 → …



Each switch:

- Floods the frame repeatedly

This is the danger:

- Unidirectional link + STP = loop

◆ **What should have happened**

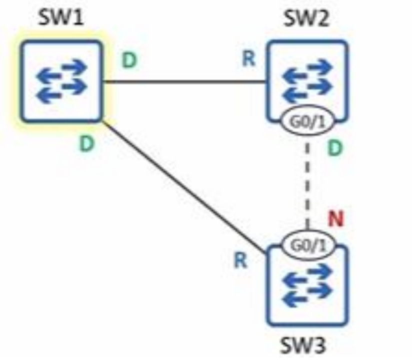
If hardware detected the issue:

- Link would go down
- Ports would disable
- No loop

But because it wasn't detected:

- STP made the wrong decision

🔹 **How Loop Guard solves this**

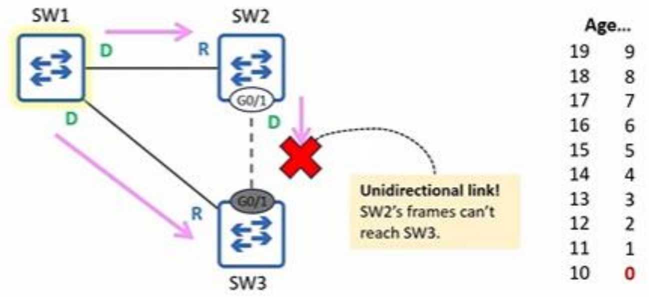


Loop Guard:

- Does **not** prevent unidirectional links
- Prevents the **STP failure** they cause

Behaviour:

- If a Loop Guard–enabled port stops receiving BPDUs
- And Max Age timer reaches 0



Instead of:

- Becoming designated
- Entering forwarding

It enters:

- **Broken (loop inconsistent) state**

Loop Guard behaviour

Broken (loop inconsistent) state:

- Port is blocked by STP
- Port remains **up/up**
- Frames are not forwarded

This is similar to:

- Root Guard's broken (root inconsistent) state

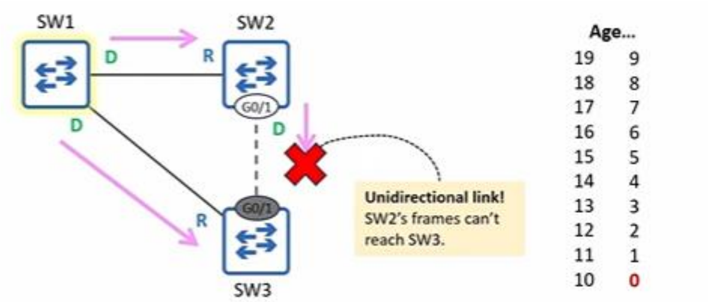
Loop Guard example

Assume:

- Loop Guard enabled on SW3 G0/1
- SW2–SW3 link is unidirectional

SW2 ========> SW3

SW2 <===X==== SW3



SW3 G0/1:

- Stops receiving BPDUs
- Max Age timer counts down

Without Loop Guard:

- Port would forward
- Loop would form

With Loop Guard:

- Port becomes **broken**
- Loop is prevented

◆ **Recovery with Loop Guard**

Recovery is **automatic**.

If:

- Physical issue is fixed
- BPDUs start arriving again

Then:

- Port automatically re-enters normal STP operation
- No manual action required

◆ **Configuring Loop Guard**

Loop Guard can be enabled in two ways:

Per-port (interface mode)

“SPANNING-TREE GUARD LOOP”

Global (default)

“SPANNING-TREE LOOPGUARD DEFAULT”

This enables Loop Guard on all ports by default.

To disable on a specific port:

“SPANNING-TREE GUARD NONE”

◆ **Where Loop Guard should be used**

Loop Guard should typically be enabled on:

- Root ports
- Non-designated ports

In other words:

- Ports that are supposed to receive BPDUs

If they stop receiving BPDUs:

- Loop Guard blocks them

- Prevents loops

◆ **Loop Guard (CLI & behaviour)**

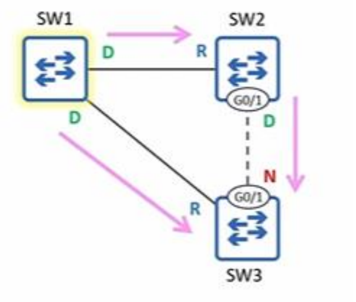
Loop Guard in the CLI

Now let’s take a look at Loop Guard in the CLI.

Loop Guard is enabled on SW3 G0/1 with:

“SPANNING-TREE GUARD LOOP”

Topology before any problem:



SW3 G0/1:

- Is receiving BPDUs from SW2
- Is operating normally

◆ **CLI (image shown once)**

SW3(config)# interface g0/1

SW3(config-if)# spanning-tree guard loop

SW3(config-if)# do show spanning-tree interface g0/1 detail

```
SW3(config)# interface g0/1
SW3(config-if)# spanning-tree guard loop
SW3(config-if)# do show spanning-tree interface g0/1 detail
 Port 2 (GigabitEthernet0/1) of VLAN0001 is alternate blocking
   Port path cost 4, Port priority 128, Port Identifier 128.2.
   Designated root has priority 1, address 5254.001a.9d29
   Designated bridge has priority 28673, address 5254.0019.f184
   Designated port id is 128.2, designated path cost 4
   Timers: message age 3, forward delay 0, hold 0
   Number of transitions to forwarding state: 0
   Link type is point-to-point by default
   Loop guard is enabled on the port
   BPDU: sent 0, received 359
```

Observed:

- “Loop guard is enabled on the port”

That’s it:

- Only **one command** is needed to enable Loop Guard on a port

◆ **Enabling Loop Guard by default**

Instead of interface config mode:

- Loop Guard can be enabled in global config mode

Command:

“SPANNING-TREE LOOPGUARD DEFAULT”

```
SW3(config)# spanning-tree loopguard default
SW3(config)# do show spanning-tree interface g0/1 detail
 Port 2 (GigabitEthernet0/1) of VLAN0001 is alternate blocking
   Port path cost 4, Port priority 128, Port Identifier 128.2.
   Designated root has priority 1, address 5254.001a.9d29
   Designated bridge has priority 28673, address 5254.0019.f184
   Designated port id is 128.2, designated path cost 4
   Timers: message age 3, forward delay 0, hold 0
   Number of transitions to forwarding state: 0
   Link type is point-to-point by default
   Loop guard is enabled on the port by default
   BPDU: sent 0, received 359
```

Verification output:

- Same as before
- But it says **“by default”** at the end

◆ **Loop Guard blocking a port**

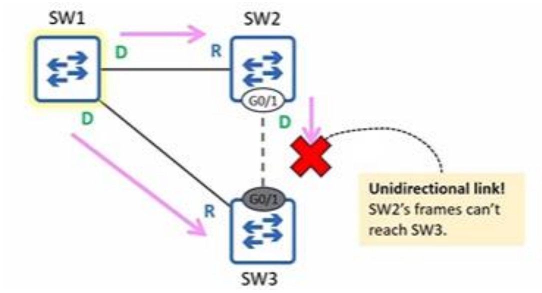
At first:

- All links are operational
- SW3 G0/1 is receiving BPDUs from SW2

Unidirectional link occurs

SW2 ========> SW3

SW2 <===X==== SW3



SW2’s frames:

- Can’t reach SW3

CLI message

- Loop guard blocking port GigabitEthernet0/1

```
%SPANTREE-2-LOOPGUARD_BLOCK: Loop guard blocking port GigabitEthernet0/1 on VLAN0001.
```

SHOW SPANNING-TREE output

```
SW3(config-if)# do show spanning-tree

!output omitted

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ----------------
Gi0/0               Root FWD 4         128.1    P2p
Gi0/1               Desg BKN*4         128.2    P2p *LOOP_Inc
```

BKN = Broken
LOOP_Inc = Loop Inconsistent

Meaning:

- **BKN** = Broken
- **LOOP_Inc** = Loop Inconsistent

Explanation:

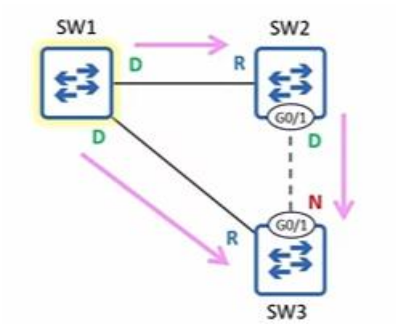
- The port stopped receiving BPDUs
- Loop Guard blocked the port
- The port is blocked by STP, not shut down

Possible cause:

- Sharp bend in a fiber
- Degraded signal
- SW2’s BPDUs don’t reach SW3

Automatic recovery with Loop Guard

Later:



- Someone adjusts the cable
- Data starts flowing again
- BPDUs from SW2 reach SW3

CLI message

- Loop guard unblocking port GigabitEthernet0/1

```
%SPANTREE-2-LOOPGUARD_UNBLOCK: Loop guard unblocking port GigabitEthernet0/1 on VLAN0001.
```

SHOW SPANNING-TREE output

```
SW3(config-if)# do show spanning-tree

!output omitted

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/0               Root FWD 4         128.1    P2p
Gi0/1               Altn BLK 4         128.2    P2p
```

Observed:

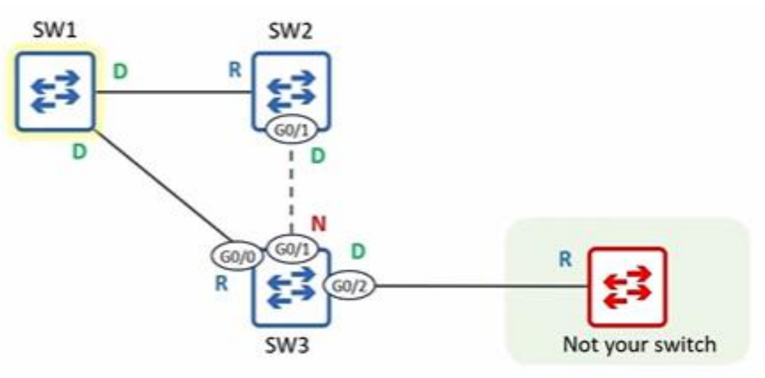
- Status is now **blocking**, not broken
- LOOP_Inc is gone

Key point:

- Loop Guard automatically re-enables the port
- No commands are needed to re-enable it

◆ Loop Guard and Root Guard together

Another switch is added:



- A switch that is **not your switch**
- You don't control its configuration

Important rule

Loop Guard and Root Guard are mutually exclusive

They:

- Cannot be enabled on the same port at the same time
- Serve opposite purposes

◆ Difference between Root Guard and Loop Guard

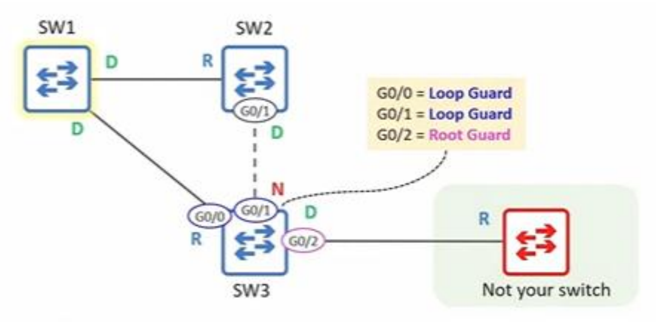
Root Guard

- Applied to Designated ports
- Prevents them from becoming Root ports
- Blocks the port if superior BPDUs are received

Loop Guard

- Applied to Root ports or Non-Designated ports
- Prevents them from becoming Designated ports
- Blocks the port if BPDUs stop arriving

◆ **Correct usage on SW3**



On SW3:

- **G0/0 → Loop Guard**
- **G0/1 → Loop Guard**
- **G0/2 → Root Guard**

Why:

- G0/0 and G0/1 should receive BPDUs
- G0/2 connects to a switch you don't control

Wrong configuration example:

- If Root Guard is configured on G0/0 or G0/1
- Those ports would be blocked
- Communication would be cut off

◆ **Configuration precedence (important)**

If:

- Loop Guard is configured on a port
- And then Root Guard is configured

Result:

- Loop Guard is disabled
- Root Guard replaces it

If:

- Root Guard is configured

- And then Loop Guard is configured

Result:

- Root Guard is disabled
- Loop Guard replaces it

If:

- Loop Guard is enabled by default
- Root Guard is configured on an interface

Result:

- Loop Guard is disabled on that interface

Key rule:

- **More specific command wins**
- Interface-level command overrides global command

This is also true for:

- PortFast
- BPDU Guard
- BPDU Filter

- **Loop Guard summary**

- Loop Guard protects the network if a port **unexpectedly stops receiving BPDUs**
- This can be caused by:
  - Software bugs
  - Hardware issues
  - Unidirectional links
- A unidirectional link:
  - Works in only one direction
  - Is usually caused by Layer 1 issues
  - Is most common with fiber-optic cables
- If a root or non-designated port stops receiving BPDUs:
  - It may become designated
  - This can cause a Layer 2 loop
- Loop Guard prevents this by:
  - Putting the port into **broken (loop inconsistent)** state
- Recovery:
  - Automatic
  - Happens when BPDUs are received again
- Configuration methods:
  - Per-port:
  - SPANNING-TREE GUARD LOOP
  - Global:
  - SPANNING-TREE LOOPGUARD DEFAULT
- To disable on a port:
- SPANNING-TREE GUARD NONE
- Final point:
  - Loop Guard and Root Guard **cannot be active on the same port**
  - Only one can be enabled at a time